

Muhammad Alyan Asmat

alyanasmat13@gmail.com | [Website Portfolio](#) | linkedin.com/in/muhammad-alyan-asmat | github.com/alyanasmat13

EDUCATION

University Of Houston

Bachelor of Science, Computer Science

Houston, TX

May 2026

EXPERIENCE

Lead Instructor

Coder Kids

March 2023 – July 2023

Houston, TX

- Designed and delivered programming curriculum to classes of up to 20 students aged 5-10, adapting technical concepts in Scratch and Python across varying skill levels and learning styles
- Led hands-on game development projects using pygame, breaking down control flow and syntax into structured exercises that progressed students from zero to functional programs
- Mentored students through debugging and problem-solving challenges, building independent thinking habits rather than providing direct answers

PROJECTS

One Piece Codenames | *React, TypeScript, Node/Express, Socket.io*

May 2026

- Built a real-time multiplayer word game supporting concurrent players across separate devices, engineered around a server-authoritative architecture to act as the single source of truth for game state.
- Solved the hidden information problem by implementing server-side role-based state filtering, ensuring spymasters and operatives receive different views of the same game without client-side data leaking.
- Curated and structured a custom JSON word list of 220+ One Piece terms organized by category (characters, locations, devil fruits, and more) to replace the standard word set.
- Deployed full-stack application to Render with persistent room management and real-time synchronization across all connected clients.

Pokémon Classifier | *Python, FastAPI, pandas, NumPy, Uvicorn*

July 2026

- Fine-tuned a Hugging Face DistilBERT transformer model to classify free-text natural language descriptions into 151 distinct classes, achieving an impressive 79% validation accuracy and a 0.82 weighted F1-score across 1,301 test samples.
- Engineered an automated data pipeline integrating with external REST APIs to systematically construct a high-quality, custom dataset containing 6,501 unique text descriptions, effectively eliminating contradictory training examples.
- Designed, built, and deployed a real-time, low-latency machine learning inference endpoint using FastAPI and Uvicorn to serve the fine-tuned transformer model for on-demand text classification..

GuessPKMN | *React, TypeScript, TailwindCSS, Firebase*

September 2025

- Built a full-stack Pokémon guessing game by architecting a custom backend service that normalizes and re-exposes PokeAPI data, decoupling the client from an unstable external dependency.
- Solved a data freshness and performance problem by designing a rolling 50-entity randomization cache with generation-based filtering to enforce uniqueness constraints without repeated API calls.
- Implemented Firebase authentication and a real-time database schema to persist and update user lifetime high scores across sessions.

TECHNICAL SKILLS

Languages: C++, Python, Java, JavaScript, TypeScript, HTML/CSS

Frameworks/Libraries: React.js, Next.js, Node, Express.js, TailwindCSS, Jest, Socket.io, FastAPI, NumPy, Uvicorn, pandas

Tools/Technologies: Git, GitHub, MySQL, Postman, Cloudflare, VSCode, Postgres, Figma, Docker, Render, Supabase, Firebase